

Wave CPG model for autonomous decentralized multi-legged robot: Gait generation and walking speed control

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Abstract

In this paper, we propose a method to control gait generation and walking speed control for an autonomous decentralized multi-legged robot by using a wave Central Pattern Generator (CPG) model. The wave CPG model is a mathematical model of nonlinear oscillators and generates rhythmic movements of the legs. The gait generation and the walking speed control are achieved by controlling the virtual energy of the oscillators (Hamiltonian). A real robot experiment showed the relationship to the Hamiltonian, the actual energy consumption and the walking speed, and the effectiveness of the proposed method was verified.

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1. Introduction

Legged animals have the capability of generating gait patterns suitable to their walking speeds. The gait-generating system consists of a Central Pattern Generator (CPG) that consists of a group of neurons located in the central nervous system [1]. Fig. 1(a) shows the schema of a CPG in an insect. In the figure, the CPG is depicted as a system of coupled nonlinear oscillators that coordinates leg movements [2–4].

A CPG model is used to control robot locomotion, and provides the robot with pattern diversity and environment adaptability. In addition, the decentralized construction of the CPG provides abilities that animals do not have. Fig. 1(b) shows an autonomous decentralized multi-legged robot. The legs of this robot function as autonomous partial systems (subsystems), and the functions of a CPG are obtained by

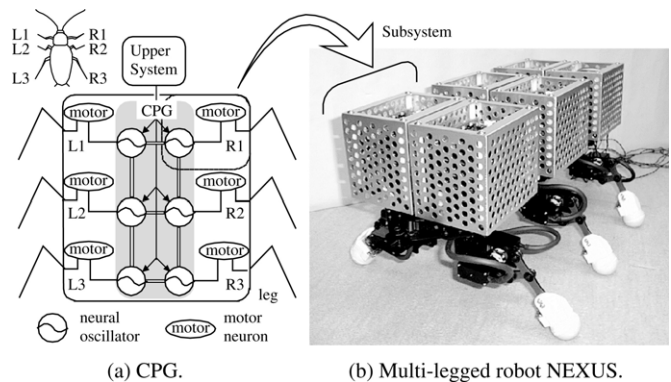


Fig. 1. CPG and multi-legged robot NEXUS.

coupling neighboring oscillators, that is, by communicating with neighboring subsystems [5,6]. Changeability of multiple legs, ease of maintenance and fault tolerance is expected from this architecture.

An autonomous decentralized multi-legged robot is a type of reconfigurable robot. Pioneering research has been carried

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out regarding the locomotion of reconfigurable robots [7–9]. However, an autonomous decentralized multi-legged robot is specialized for walking. Therefore, more efficient locomotion is also another possibility.

Regarding autonomous decentralized multi-legged robots, Tsujita et al. proposed a CPG model for which a basic gait is adapted to the walk speed [6]. Odashima et al. proposed a hierarchical structure for an upper system to control the subsystems [10]. To realize more efficient locomotion, it is important to generate gait patterns suitable to both the walking speed and the number of legs.

At the time of previous research, we developed a CPG model that can adapt to changes in the number of oscillators [11]. The model is called the ‘wave CPG model’. The wave CPG model name is derived from the analogy between a gait and a wave that occurs in the linked oscillators. It is possible to control the gait generation and transition by controlling the virtual energy of the oscillators. In this paper, we improve and utilize this model to control the leg movements of an autonomous decentralized multi-legged robot. We focus especially on how the legs decide movements from the local information. In addition, we evaluate qualitatively the proposed method from experimental results. While we discuss the changeability of multiple legs, we do not deal with fault tolerance in this paper.

2. Outline of the proposed system

The proposed system is summarized in Fig. 2. A target value for the virtual energy of an oscillator is represented as H_T , which is input to each oscillator. The virtual energy of an oscillator is called a ‘local Hamiltonian value’, and the sum of all local Hamiltonian values is called ‘total Hamiltonian value’. H_T is a target value for a local Hamiltonian value. In this research, we permit the upper system to be centralized. The upper system means an operator or a controller in a higher control layer.

Each oscillator has two variables: the radius of the limit cycle $r(u)$ and a phase angle $\theta(u)$, where u is an index of oscillators. The wave pattern on the linked oscillators converges to an eigen-mode when the local Hamiltonian values converge to H_T . Consequently, the parameters of the oscillators are independent of the total number of oscillators. A toe trajectory generator transforms the oscillator variables $r(u)$ and $\theta(u)$ to a target position of the toe $(x_d(u), y_d(u), z_d(u))$. Each leg movement is PD-controlled. As the result, the open-loop control of the gait and walking speed is achieved by means of the target value of the local Hamiltonian H_T .

3. Wave CPG model

3.1. The characteristics of the wave CPG model

We constructed a wave CPG model based on our previous research [11].² Fig. 3 shows the linking of the oscillators

² In Ref. [8], however, the oscillator variables are represented as the positions in the phase space $(q(u), p(u))$. In this paper, we use $r(u), \theta(u)$ as $q(u) = r(u) \cos \theta(u)$, $p(u) = r(u) \sin \theta(u)$, by which the calculation error is reduced. As a result, it becomes easy to port the wave CPG model to a microcomputer.

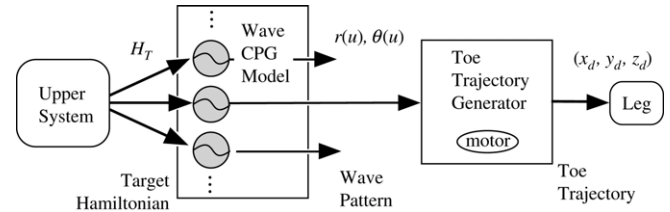


Fig. 2. System of each leg.

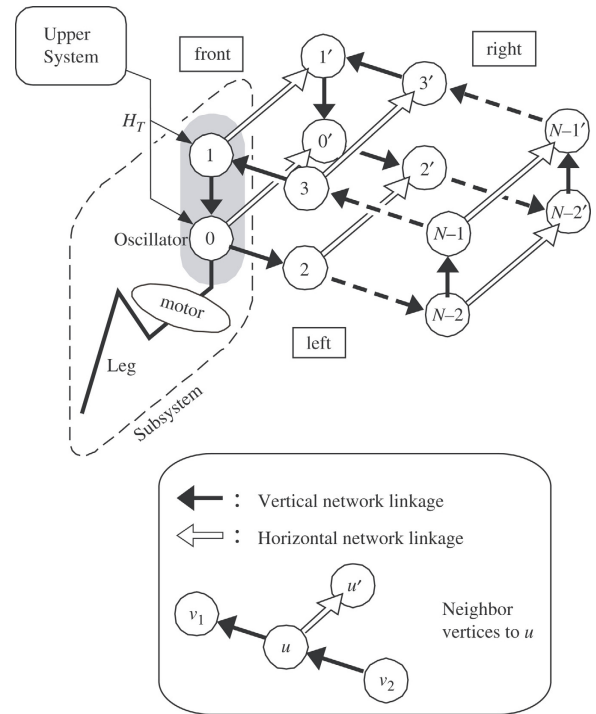


Fig. 3. Wave CPG model in N -legged robot. (The line of H_T is abbreviated.)

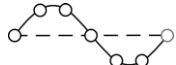
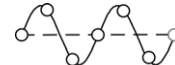
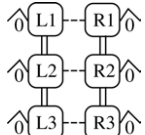
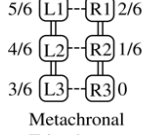
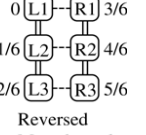
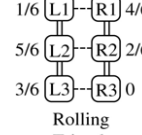
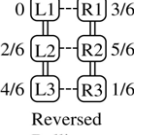
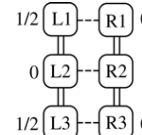
in a wave CPG model. Each leg has two oscillators and the even-numbered oscillator generates the rhythm of the leg movement. A higher system supplies the target value for the local Hamiltonian value H_T equally to all oscillators.

An N -legged robot has N -oscillator vertical networks on the left and right sides, as shown in Fig. 3 (black arrows). N is an even number. Waves are generated in both left and right vertical networks. When one of the two oscillators in a leg is used for the toe control, the legs move like a crawler. In contrast, laterally symmetrical oscillators are connected in the left and right networks (white arrows), and generate the waves on the networks anti-phase. These connections realize the properties that are generally observed in insect gaits: propagation of the swing legs from the tail to the front or vice versa, and anti-phased movement of the laterally symmetrical (left and right) legs [12].

Each oscillator only needs information on the neighboring oscillators. When a leg has two oscillators, as shown in Fig. 3, the leg can drive the oscillators using the information on the neighboring legs.

Let a set of vertices on one side of a vertical network be V and a set of edges be E , and those on the other side be V' and

Table 1
Eigen-Hamiltonian and natural modes

m	0	1, 2	3, 4	5
H_m	0	1	3	4
Mode of vertical network	 Osc. 0 2 4 5 3 1 (0) Mode 0	 Mode 1-2	 Mode 3-4	 Mode 5
Gait	 Standing	 Metachronal Tripod  Reversed Metachronal Tripod	 Rolling Tripod  Reversed Rolling Tripod	 Tripod

E' , respectively. On each vertex $u \in V$, two variables $r(u) \in R$ and $\theta(u) \in R$ are defined, as are $r(u')$ and $\theta(u')$ for $u' \in V'$. $r(u)$ is a radius and $\theta(u)$ is a phase angle such that $r(u) \geq 0$, $-\pi < \theta(u) \leq \pi$.

Given the above definitions, the dynamics of a vertex (oscillator) u are represented as

$$\frac{dr(u)}{dt} = \frac{1}{h} \sum_{v \sim u} r(v) \sin \delta(u, v) - \frac{\partial P}{\partial r(u)}, \quad (1)$$

$$\frac{d\theta(u)}{dt} = -\frac{1}{h} \left(2 - \sum_{v \sim u} \frac{r(v)}{r(u)} \cos \delta(u, v) \right) - \frac{\partial P}{\partial \theta(u)}, \quad (2)$$

where h is a positive constant, $v \sim u$ is a set of neighboring vertices to vertex u in the vertical network, $\delta(u, v) \equiv (\theta(u) - \theta(v))$, and P is a potential function defined below (Section 3.3). We named the first term and the second term of Eqs. (1) and (2) the ‘conservative term’ and ‘dissipative term’, respectively.

3.2. Conservation and the Hamilton system

The conservative terms in Eqs. (1) and (2) are derived from a wave equation. If there are no dissipative terms in Eqs. (1) and (2), the dynamics preserve the Hamiltonian value.

The Hamiltonian value is defined as follows:

$$H = \sum_{u \in V} H(u), \quad (3)$$

$$H(u) = r^2(u) \left(2 - \sum_{v \sim u} \frac{r(v)}{r(u)} \cos \delta(u, v) \right), \quad (4)$$

where $\sum_{u \in V}$ is designated as the sum for all vertices $u \in V$. H is the total energy of the oscillators and $H(u)$ is the energy of each oscillator. We refer to H as the ‘total Hamiltonian value’ and $H(u)$ as the ‘local Hamiltonian value’.

Waves generated by the conservative terms are composed from some natural oscillation modes. For example, a 6-oscillator network has six modes, as shown in Table 1. In Table 1, m is the mode number and H_m is the eigen-Hamiltonian value. The natural angular velocity ω_m is generated in proportion to H_m as per

$$\omega_m = H_m/h. \quad (5)$$

These wave patterns are obtained from solving an eigenvalue problem. Mode 5 generates a standing wave and Mode 0 is a stationary state. Modes 1 and 2, 3 and 4 have the same eigenvalues, and form gaits by switching themselves through one oscillation cycle. These oscillation patterns are referred to as ‘exchange-patterns’, e.g., ‘exchange-pattern 1 and 2’ or ‘exchange-pattern 3 and 4’. As for the wave CPG model (Fig. 3), the natural oscillation modes are matched with hexapodal gaits, as shown in Table 1.

Generally, an N -oscillator vertical network has N modes; Mode 0 is stationary and Mode $N - 1$ generates a standing wave, and Modes 1 to $N - 2$ generate traveling waves with two modes. When a standing wave or an exchange-pattern appear and $r(u) = 1$ for all $u \in V$, then the local Hamiltonian value $H(u)$ is expressed as

$$H(u) = 2 - 2 \cos \frac{2n\pi}{N}. \quad (6)$$

This equation is derived from Eq. (4), $r(u) = 1$ and the cyclic boundary conditions of the eigen-modes. Therefore, the target value of the local Hamiltonian value is selected from $[0, 4]$ regardless of the number of oscillators.

3.3. Dissipation and potential functions

The dissipative terms in Eqs. (1) and (2) work to minimize the potential function P . The potential functions P is set, as follows,

$$P = \sum_{u \in V} (P_0(u) + P_1(u) + P_2(u)), \quad (7)$$

$$P_0(u) = \frac{a}{2} \left(r^2(u) + \frac{1}{r^2(u)} \right), \quad (8)$$

$$P_1(u) = \frac{b}{2} (H(u) - H_T)^2, \quad (9)$$

$$P_2(u) = \frac{c}{2} (r(u) - r(u'))^2 + \frac{d}{2} \sum_{u \in V} \{ (\sin \delta \theta(u, u'))^2 + (\cos \delta \theta(u, u') + 1)^2 \}, \quad (10)$$

where a , b , c and d are time constants that represent the speeds of convergence, and H_T is a target value of the local

Hamiltonian value. $P_0(u)$ makes the variable of each vertex converge to a unit circle in the phase plane. $P_1(u)$ makes the local Hamiltonian $H(u)$ of each vertex follow the target value H_T . $P_2(u)$ makes waves in the left and right networks become anti-phase.

The dissipative terms of $P_0(u)$ and $P_1(u)$ make the wave pattern converge to a natural oscillation mode with $r(u) = 1$. In addition, when the target value H_T of local Hamiltonian $H(u)$ increases (or decreases) continuously, the variable of each vertex follows it by enlarging (or lessening) the orbital radius, at first. When the amount of the change in the radius becomes even larger, the variables of the vertices change (bifurcate) the natural oscillation mode, and continue the follow-up to the target value H_T .

The dissipative terms in Eqs. (1) and (2) are calculated as follows,

$$\begin{aligned} \frac{\partial P}{\partial r(u)} = & a \left(r(u) - \frac{1}{r^3(u)} \right) \\ & + b \left\{ 4(H(u) - H_T)r(u) - \sum_{v \sim u} (H(u) \right. \\ & + H(v) - 2H_T)r(v) \cos \delta\theta(u, v) \left. \right\} \\ & - 2c(r(u) - r(u')), \end{aligned} \quad (11)$$

$$\begin{aligned} \frac{\partial P}{\partial \theta(u)} = & b \sum_{v \sim u} (H(u) + H(v) - 2H_T)r(u)r(v) \sin \delta\theta(u, v) \\ & - d(\cos \theta(u) \sin \theta(u') - \sin \theta(u) \cos \theta(u')). \end{aligned} \quad (12)$$

Eqs. (1), (2), (11) and (12) show that the dynamics at each vertex are determined by the values of neighboring vertices $r(v)$, $\theta(v)$, $H(v)$, $r(u')$, $\theta(u')$.³

4. Toe trajectory generator

In this section, the variables of an oscillator $r(u)$ and $\theta(u)$ are mapped to the target trajectory of the toe movement ($x_d(u)$, $y_d(u)$, $z_d(u)$). In this paper, we deal with the leg shown in Fig. 4. Motor 1 produces propulsive force and Motors 2 and 3 produce standing force. Each leg has a local coordinate system, but note that the y-axis is the direction in which the robot walks. That is, the left-side legs and the right-side legs have a left-handed system or a right-handed system, respectively. Therefore, each leg needs information about whether it is on the left side or the right side. In the following discussion, we assume that $\omega(u) < 0$. This assumption is derived easily when the dissipative term in Eq. (2) is not much greater than the conservative term.

4.1. Toe trajectory in swing phase

In the swing phase of the leg movement, the leg is controlled so that the duration of the swing phase becomes constant. The

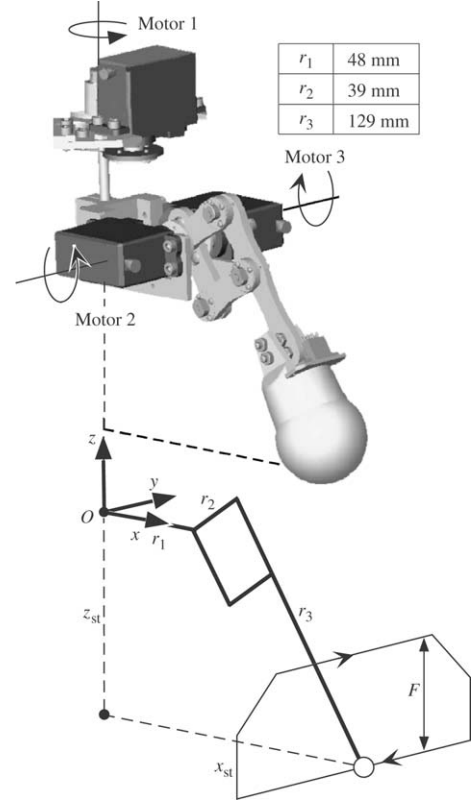


Fig. 4. Leg of robot.

duration of the swing phase depends on the motor's maximum speed or the upper limit of the duration of the swing phase. The limit of the duration is discussed in Section 4.3. The target trajectory of a toe is shown in Fig. 6 (left). The swing phase starts at $\theta(u) = \pi$, and after T_{sw} [s] the phase changes to the stance phase. The step length $S(u)$ is in proportion to $r(u)$ as follows,

$$S(u) = S_o r(u), \quad (13)$$

where S_o is the default step length. The toe trajectory is

$$x_d(u) = x_{st} \quad (14)$$

$$y_d(u) = \begin{cases} -S(u)/2 & 0 \leq t - t_{sw} < T_{sw1} \\ S(u) \left(-\frac{1}{2} + \frac{t - t_{sw} - T_{sw1}}{T_{sw4} - T_{sw1}} \right) & T_{sw1} \leq t - t_{sw} < T_{sw4} \\ S(u)/2 & T_{sw4} \leq t - t_{sw} < T_{sw} \end{cases} \quad (15)$$

$$z_d(u) = \begin{cases} z_{st} + \frac{F(t - t_{sw})}{T_{sw2}} & 0 \leq t - t_{sw} < T_{sw2} \\ z_{st} + F & T_{sw2} \leq t - t_{sw} < T_{sw3} \\ z_{st} + \frac{F\{-(t - t_{sw}) + T_{sw}\}}{T_{sw} - T_{sw3}} & T_{sw3} \leq t - t_{sw} < T_{sw} \end{cases} \quad (16)$$

where $t = [t_{sw}, t_{sw} + T_{sw}]$, and $T_{sw1} \leq T_{sw2} \leq T_{sw3} \leq T_{sw4} \leq T_{sw}$. x_{st} is the distance from the toe to the body along the x-axis. z_{st} is the height from the toe to the body along the z-axis.

4.2. Toe trajectory in the stance phase

In the stance phase, as shown in Fig. 6 (right), the toe trajectory depends on $\theta(u)$. When the swing phase changes to

³ In fact, a vertex needs information for the two next neighboring vertices, because the two next neighbors' variables are included in $H(v)$ (see Eqs. (11) and (12)). It is possible to eliminate this problem by having each subsystem receive the $H(v)$ values calculated in the neighbors.

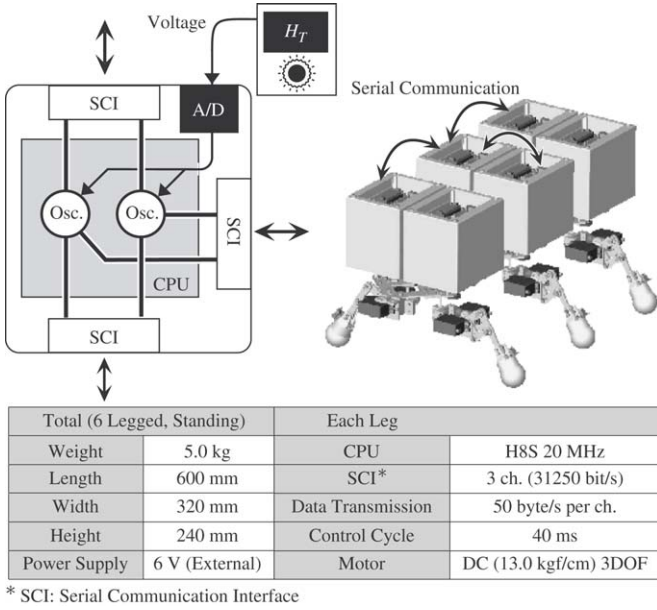


Fig. 5. Specification of NEXUS.

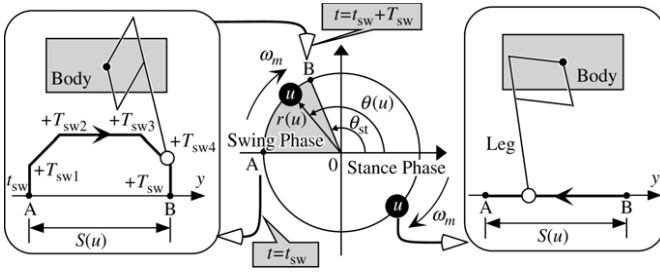


Fig. 6. Swing phase and stance phase.

the stance phase at $\theta(u) = \theta_{st}$, the range of the stance phase $\theta = [\theta_{st}, -\pi]$ is mapped to the toe trajectory. That is,

$$x_d(u, \theta(u)) = x_{st}, \quad (17)$$

$$y_d(u, \theta(u)) = S(u) \left(\frac{\theta(u) + \pi}{\theta_{st} + \pi} - \frac{1}{2} \right), \quad (18)$$

$$z_d(u, \theta(u)) = z_{st}. \quad (19)$$

There is a possibility that large internal forces may occur between the legs because the trajectory of each leg is determined identically. Therefore, the actuators should be controlled by PD-control with low stiffness.

4.3. Duty factor and lower limit of duration of the swing phase

When the angular velocity $\omega(u) \equiv \dot{\theta}(u)$ is assumed to be constant in one step, the duty factor β is calculated as

$$\beta = \frac{\pi + \theta_{st}}{2\pi} = \frac{2\pi + T_{sw}\omega(u)}{2\pi}. \quad (20)$$

The duty factor needs to be kept at more than 0.5 for static walking. Therefore, we have to set an upper limit on the duration of the swing phase. The duty factor β must be more than 0.5 even when a natural oscillation mode appears with a

maximum natural frequency. The maximum natural frequency is $h/4$ regardless of the total number of oscillators⁴ and, from Eq. (20), we obtain the upper limit as

$$T_{sw} \leq \frac{\pi h}{4}. \quad (21)$$

5. Changeability of multiple legs

It is easy for the wave CPG model to extend the number of legs. The increment and decrement in the number of legs are accomplished by remaking closed vertical networks (Fig. 7). In the same way, if the robot is severed in the middle, they will work as two independent walking robots. As far as the vertical networks are closed, it is possible to generate wave patterns as discussed in the previous sections. High exchangeability and maintainability are accomplished using this idea. However, when there are not many legs, for example only two or four, the robot is not able to walk well. In such a case, additional functions are needed to maintain stability while standing.

The robot that we used in the experiments cannot reconfigure itself autonomously. We changed the connections by hand. Autonomous reconfiguration needs a function for detecting the neighboring legs and a mechanism for fixing and releasing the neighboring legs.

6. Experiments

6.1. Autonomous decentralized multi-legged robot: NEXUS

The robot used in the experiments is shown in Fig. 5. Each subsystem has a leg and a microcomputer. The subsystems are connected mechanically and communicate with neighbors via a Serial Communication Interface (SCI). As shown in Fig. 5 (left), each microcomputer calculates the dynamics of two oscillators, and sends and receives the variables. The subsystem can sense the motors' angles and the electrical energy consumed in the motors. The target value of the local Hamiltonian value H_T is input as a common voltage value using AD conversion.

We experimented with three types of the robot: 6-legged, 8-legged, and 10-legged. In these experiments, the robot walked in a straight line on a treadmill. Parameters in Eqs. (1), (2), (11) and (12) are selected experimentally as $\{h, a, b, c, d\} = \{1.25, 2.0, 0.2, 0, 0\}$ in the 6-legged robot experiment, and as $\{h, a, b, c, d\} = \{2.5, 2.0, 0.2, 2.0, 0.25\}$ in the 8- and 10-legged robot experiments. The initial variables of $r(u)$ and $\theta(u)$ were generated using uniform random numbers for $[0, 10]$ and $[-\pi, \pi]$, respectively. In the toe trajectory generator, parameters are $S_0 = 60$ mm, $F = 18$, $x_{st} = 130$, and $z_{st} = -90$. T_{sw} was 0.86 ms in the 6-legged robot experiment and 0.96 ms in the 8- and 10-legged robot experiments.

⁴ This relation is obtained from Eq. (5) and $H_m = 4$.

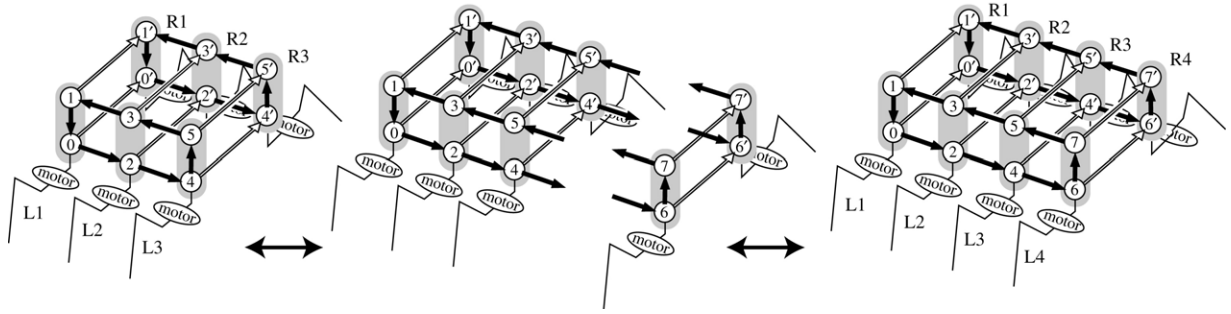


Fig. 7. Increment or decrement in the number of legs.

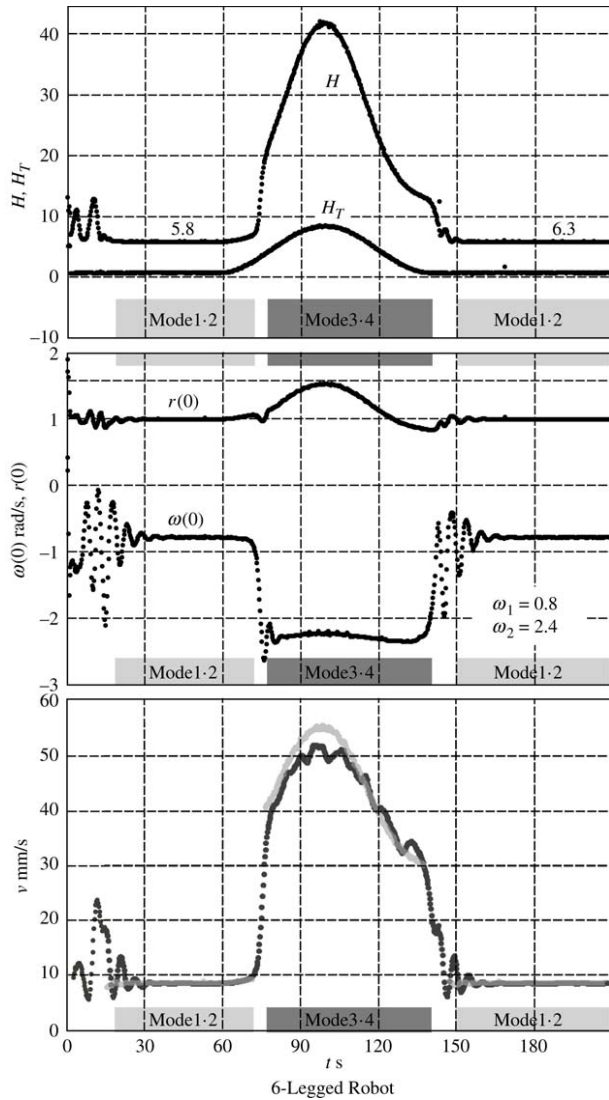


Fig. 8. Experimental data of 6-legged robot.

In the 6-legged robot experiment, the right-side legs are made to follow the left-side legs in order to decrease the communication cost. The variables for the right side are calculated as $(r(u'), \theta(u'), H(u')) = (r(u), \theta(u) + \pi, H(u))$; that is, the left-side legs do not calculate the dynamics of the oscillators. This is realized by setting the parameters c and

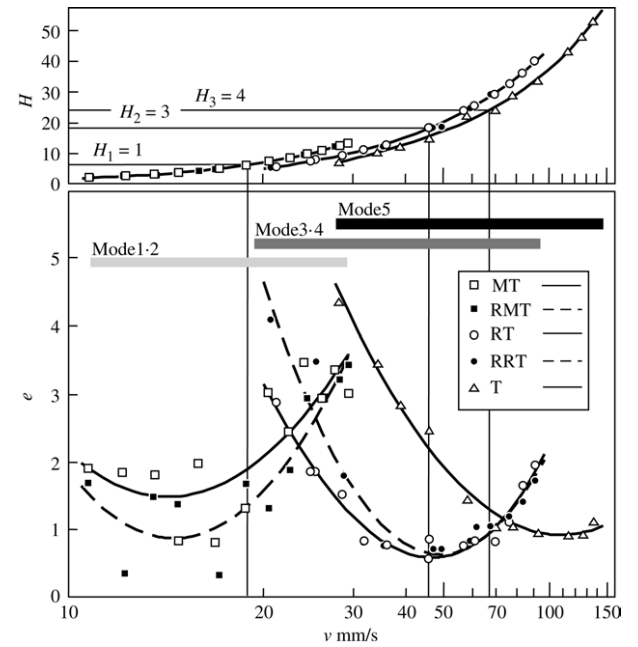


Fig. 9. Hamiltonian and energy efficiency versus walking speed (6-legged).

d to 0. As a result, calculation time of the 6-legged robot becomes shorter. In the 8- and 10-legged robot, the left-side legs calculate the dynamics and communicate with the right side.

6.2. Gaits and walking speed

Fig. 8 is the result of the 6-legged robot experiment. H_T is changed as shown in Fig. 8 (top). The total Hamiltonian value H of the left-side vertical network is also plotted in this figure. The oscillation pattern transitioned from exchange-pattern 1 and 2 to 3 and 4, and then the gaits went to the metachronal tripod (MT). In addition, the reverse transition also occurred. It is difficult to predict the length of the transient state, because it depends on the initial conditions. In other initial conditions, Mode 5 infrequently occurs instead of 3 and 4 because their eigen-Hamiltonian values are close.

Fig. 8 (middle) shows $r(0)$ and $\omega(0)$ of the left-side network. The angular velocity $\omega(0)$ almost equals the natural angular frequency ω_m .

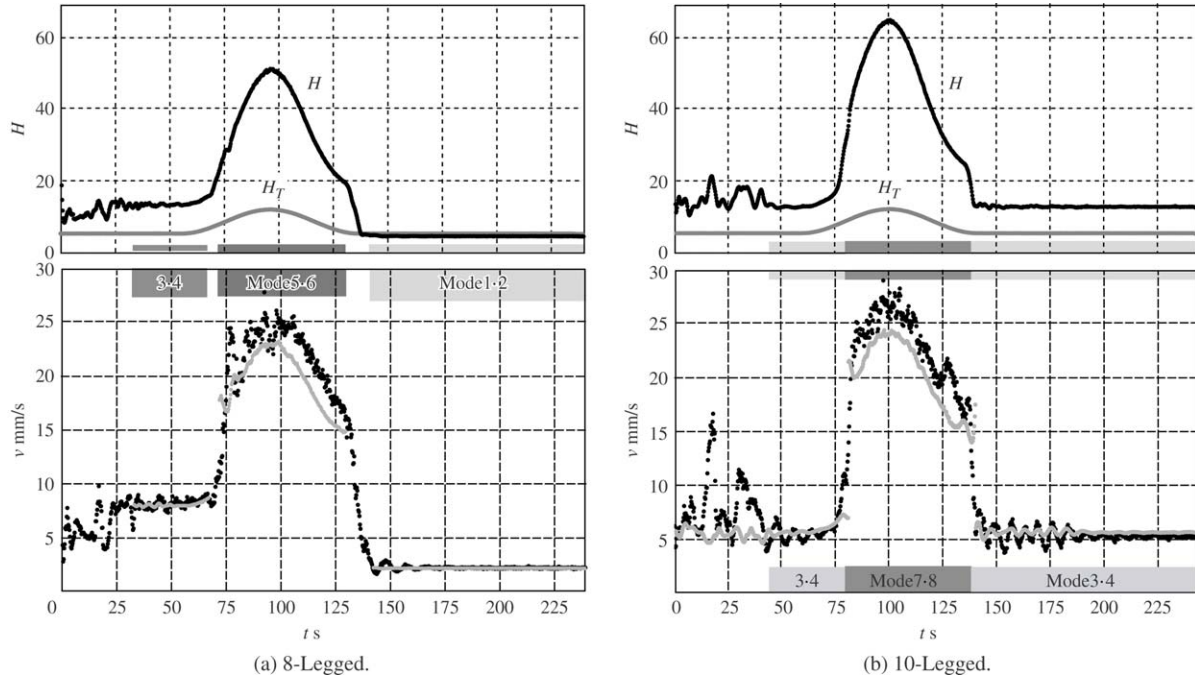


Fig. 10. Experimental data of (a) 8-legged robot and (b) 10-legged robot.

Fig. 8 (bottom) shows the walking speed v of the robot. The walking speed was calculated by using the average of the relative speeds between the standing-legs' toes and the body. The data was smoothed using a moving average, because the dispersion of the measurement error was large. The walking speed v changed dramatically at the transitions, but changed smoothly through the same gaits. As a result, the change in the walking speed corresponds to the total Hamiltonian value H .

When we assume that $r(u) = r(v)$, $\omega(u) = \omega(v)$ and $H(u) = H(v)$ for all $(u, v) \in V, V'$, and the toes do not slip on the ground, the walking speed is calculated theoretically as

$$v = \frac{S_0 r^3(u) H(u)}{2\pi h r^2(u) - T_{SW} H(u)}. \quad (22)$$

The gray line in Fig. 8 (bottom) is the theoretical walking speed.

Fig. 10 is the result of the 8-legged and 10-legged robot experiments. H_T was changed in the same way as the 6-legged robot experiment. While the 8-legged and 10-legged robots have 5 and 6 gait patterns respectively, only 3 and 2 patterns appeared. The patterns that appeared depended on the initial conditions.

Fig. 11 is a gait diagram in the 8-legged robot experiment. The gait diagram shows when the legs touched the ground. The black line represents the stance phase and the clearance represents the swing phase. The length of the stance phase changed according to the gait patterns, while the length of the swing phase was constant.

6.3. Hamiltonian value and energy consumption

Fig. 9 shows the relationship between the Hamiltonian value and the energy efficiency when the gaits of the 6-legged robot

are stable. Note that the horizontal axis of the graph is the logarithmic walking speed. Energy efficiency is calculated as a dimensionless number [13]:

$$e = \frac{E - E_S}{v M g}, \quad (23)$$

where E is the electrical energy consumed in all the motors, and $M = 5.0$ kg and $g = 9.81$ m/s² are the mass of the robot and the gravitational acceleration, respectively. Eq. (23) evaluates the net energy used for the robot's propulsion.

In Fig. 9, the velocity of each gait was changed by amplifying the default step length S_0 while $H_T = H_m$. Each point of measurement is an average of 20 walking strokes. The points $\square$ and $\blacksquare$, $\circ$ and $\bullet$ correspond to two gaits with the same eigen-Hamiltonian value, respectively (see Table 1). Quadratic curves are fitted to the points of measurement. However, the values of modes 1 and 2 are dispersive, because the walking speed v is the denominator in Eq. (23). The lower the walking speed v becomes, the larger the dispersion its measurement error becomes. Therefore, the calculational error is large when the walking speed is low. We attempted to fit the quadratic curves to the measurement points of modes 1 and 2. The differences between the curves MT and RMT, RT and RRT might also be based on the same reason.

Important results are derived from Fig. 9:

- the intersection of the energy-efficiency curves;
- the existence of a walking speed that maximizes the energy efficiency for each gait.

These results follow the experiment results of Hoyt and Taylor [14], which show that gait generation and transition are accomplished from the optimization of the energy efficiency.

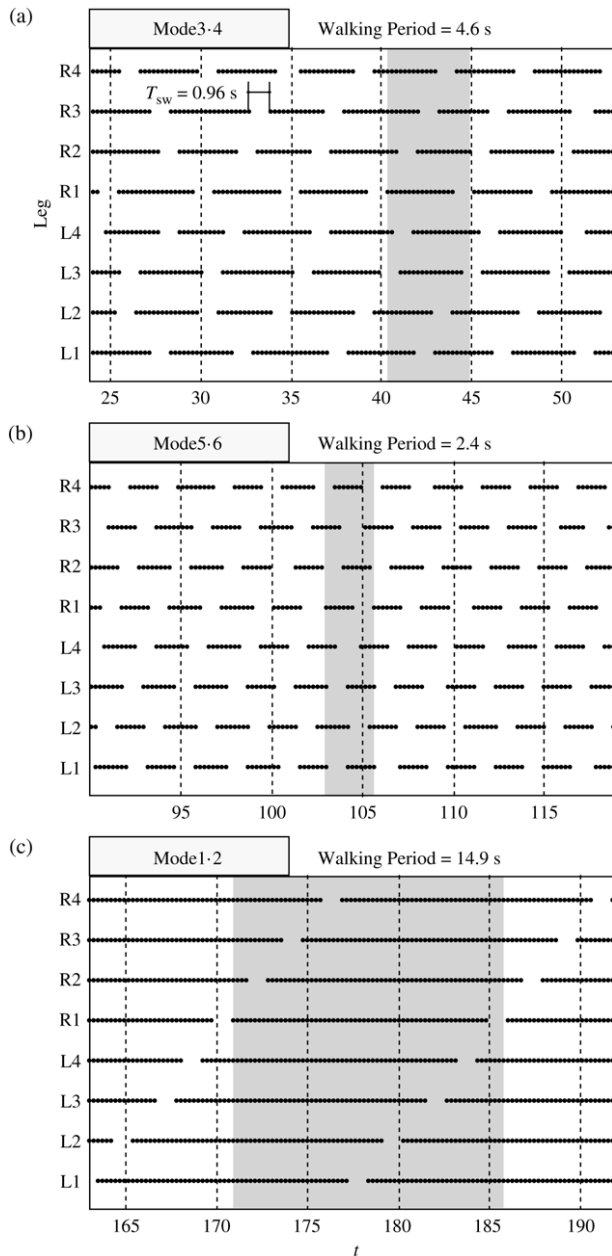


Fig. 11. Gait diagram of 8-legged robot.

The constant duration of the swing phase and the small vertical movement of the robot are considered to contribute to these results [15,16]. If the step length of each gait is selected properly, it is possible that the walking speed will jump through the intersections of the energy-efficiency curves.

7. Conclusion

In this paper, we have proposed a method concerning gait generation and speed control with a wave CPG model. The experiments revealed the possibility of controlling the walking speed and the gaits by operating the Hamiltonian value. However, the Hamiltonian value and the energy consumption do not have a direct relationship, because this method is

an open-loop system. A feedback loop from the leg to the oscillator is necessary to integrate the Hamiltonian value and the mechanical energy. If the integration is realized, we will be able to design a system in which the oscillator dynamics and the leg dynamics are integrated.

The proposed controller can be generalized as a wave pattern controller, especially for robots that have homogeneous components. It is possible for this controller to control traveling waves on a ring network, and it is also possible to make coordinated patterns on two or more ring networks via the interactions. It is for future work to propose other applications.

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